Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

GRU Multi-Step Training Report

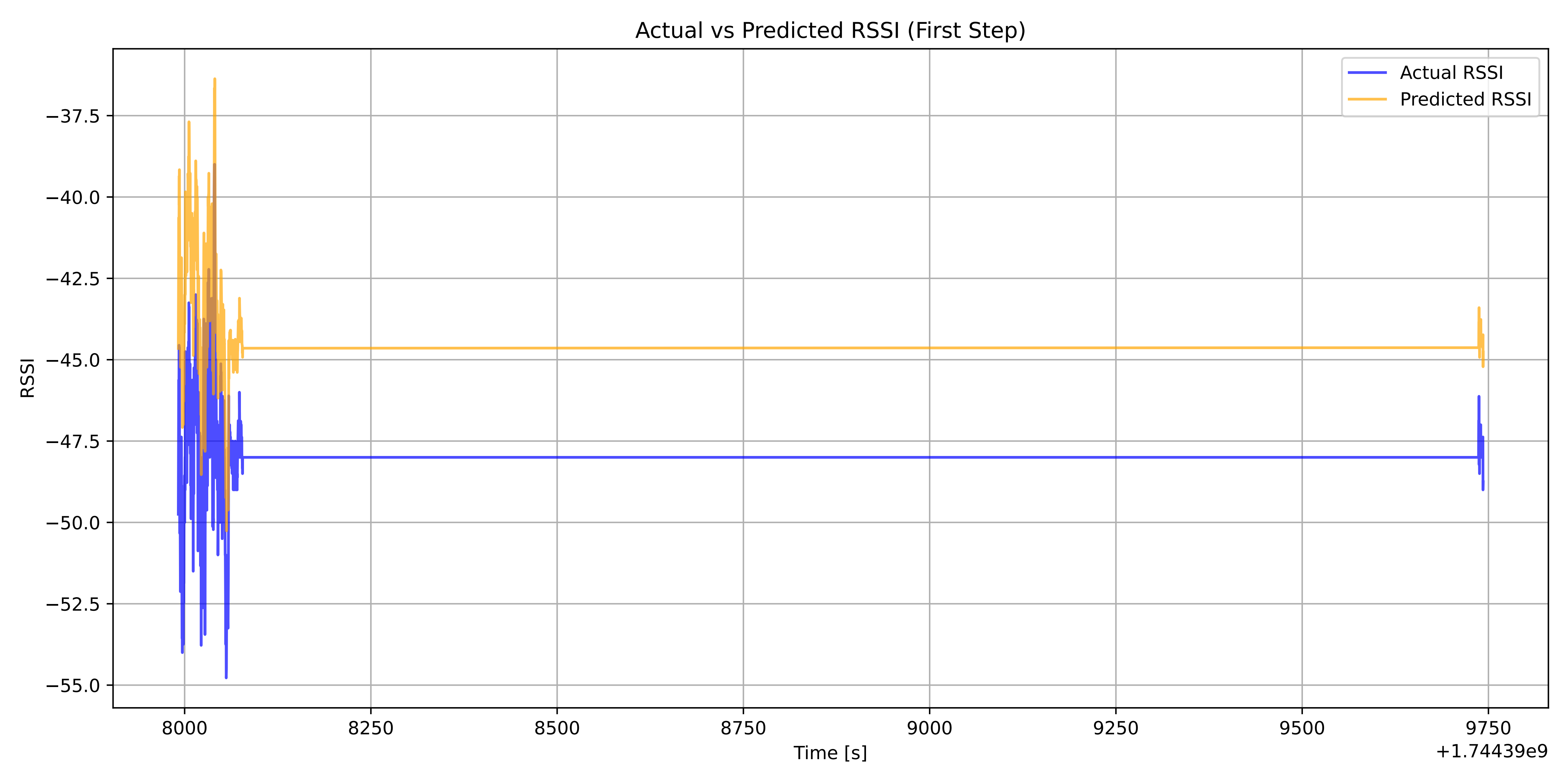
Report Generated: 2025-05-07 08:36:41

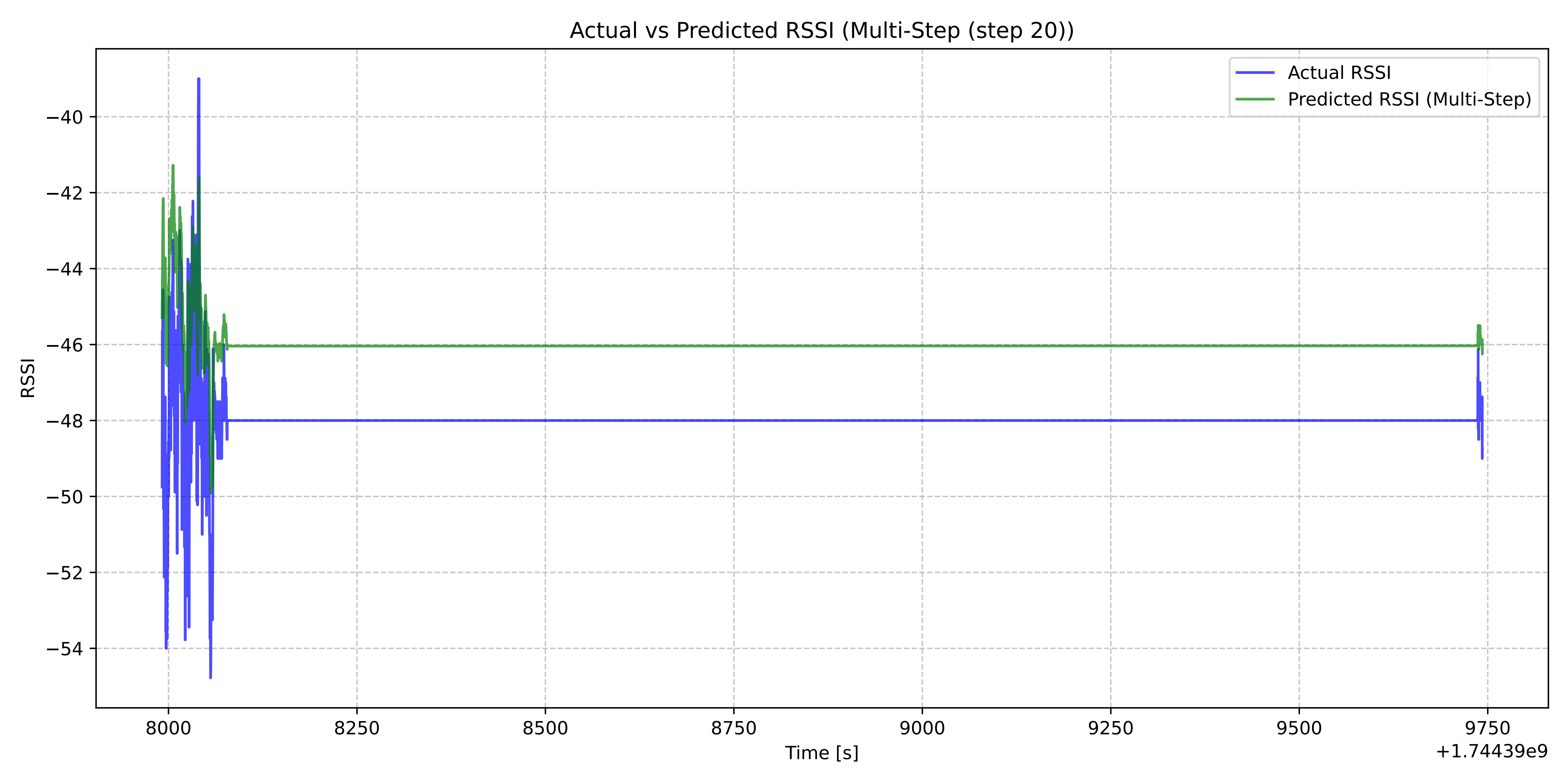
Training Start: 2025-05-06_20-00-23
Training End: 2025-05-06_20-11-25
Training Duration: 11.03 minutes

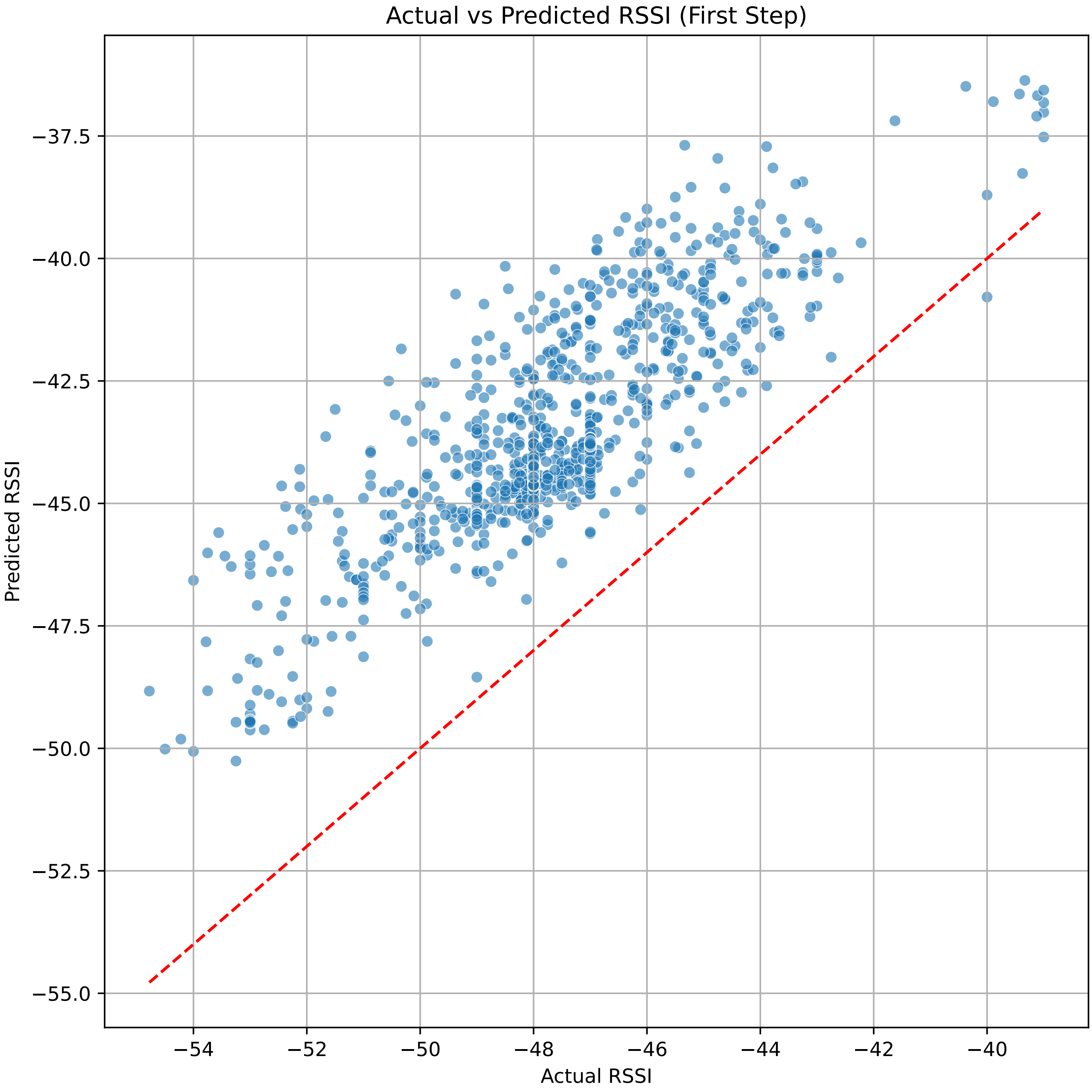
Best Hyperparameters:
Number of Layers: 2
Activation: tanh
Optimizer: Nadam
Dropout Rate: 0.20

MAE: 3.2007
MSE: 13.5430
RMSE: 3.6801
R2 Score: -1.3668

Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.1526
MSE: 0.0259
RMSE: 0.1608
R2 Score: -2.2131

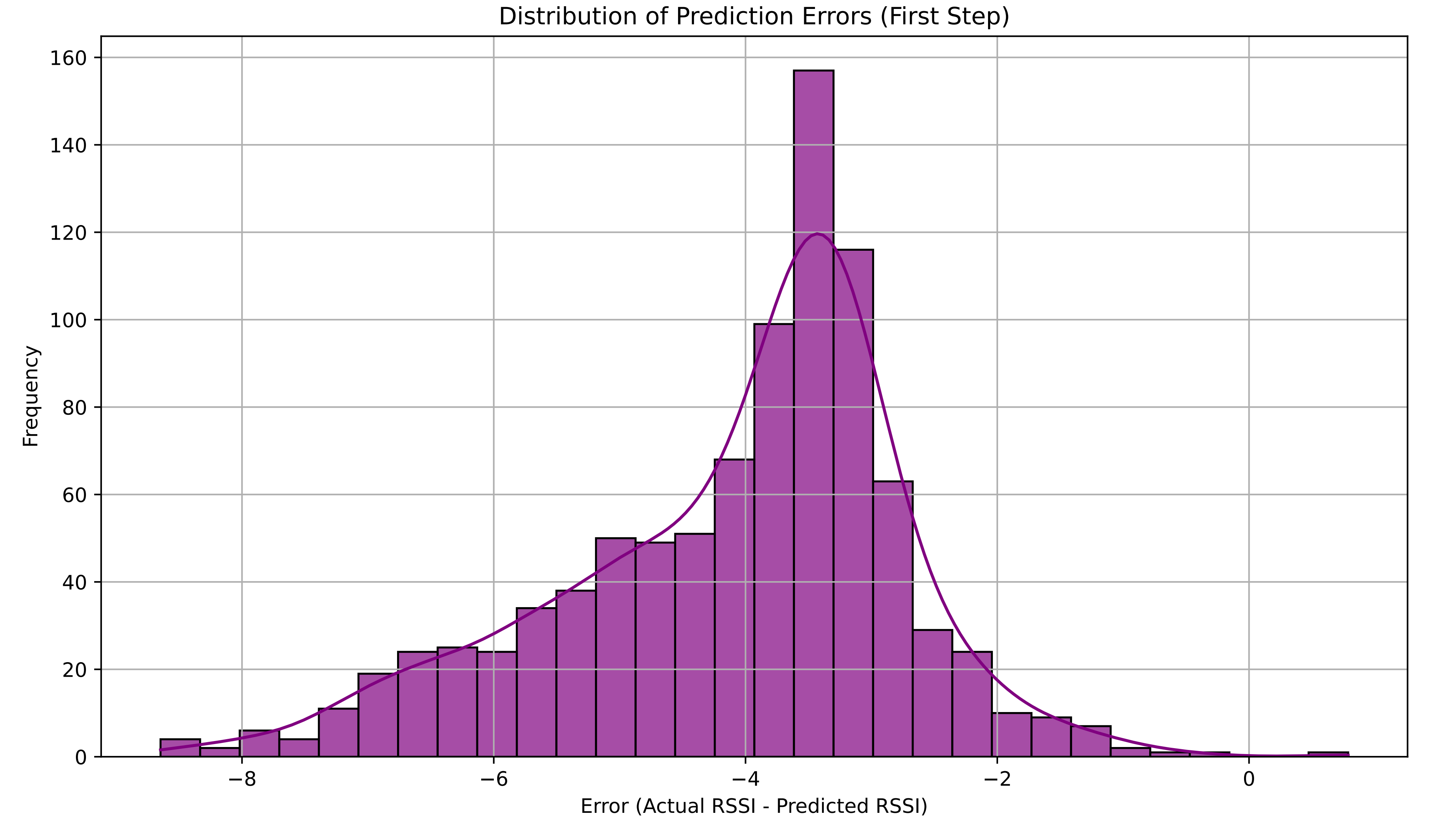

Actual vs Predicted RSSI (First Step)
Actual RSSI
Predicted RSSI
−37.5
−40.0
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
RSSI
8000
8250
8500
8750
9000
9250
9500
9750
Time [s]
+1.74439e9

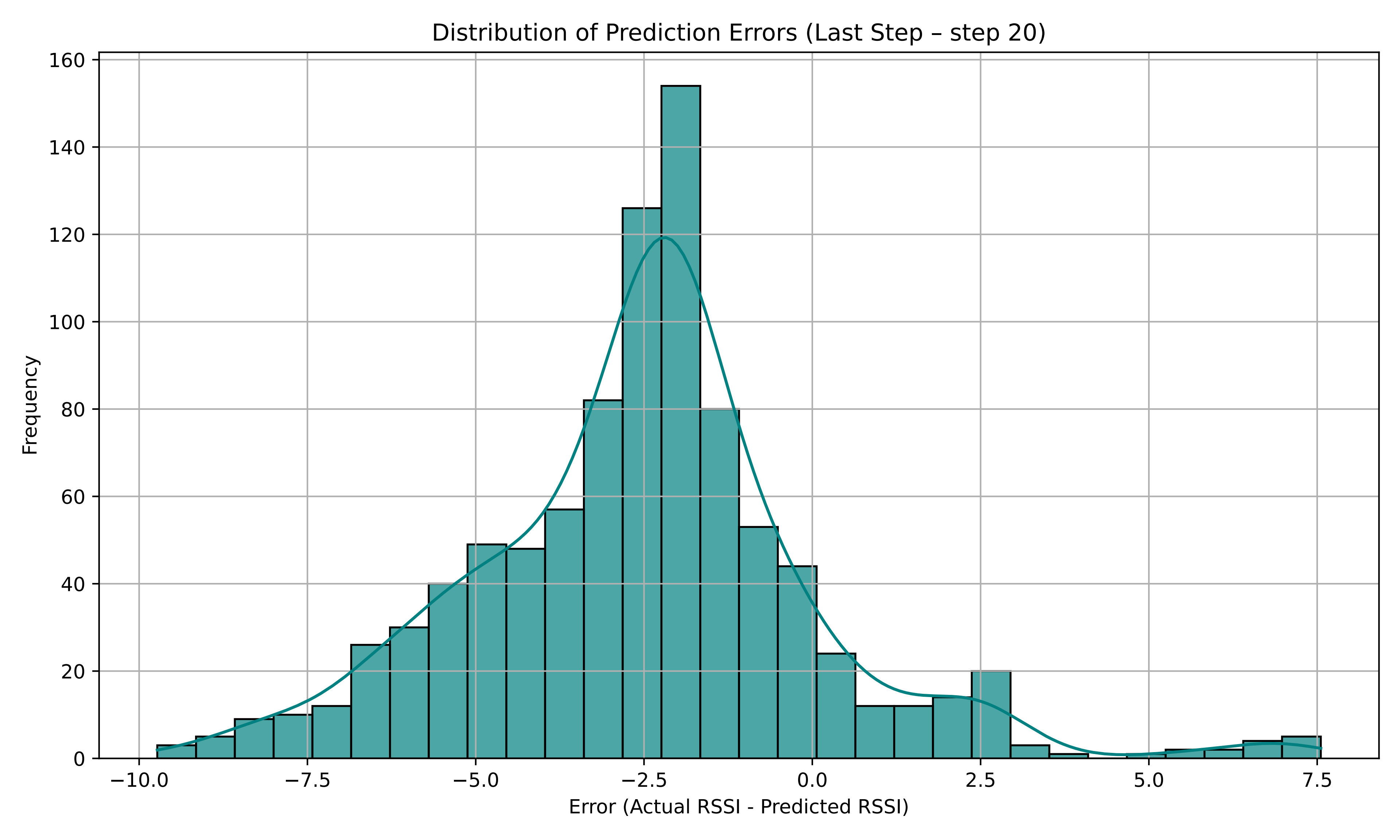

Actual vs Predicted RSSI (Multi-Step (step 20))
Actual RSSI
Predicted RSSI (Multi-Step)
RSSI
−40
−42
−44
−46
−48
−50
−52
−54
8000
8250
8500
8750
9000
9250
9500
9750
+1.74439e9
Time [s]

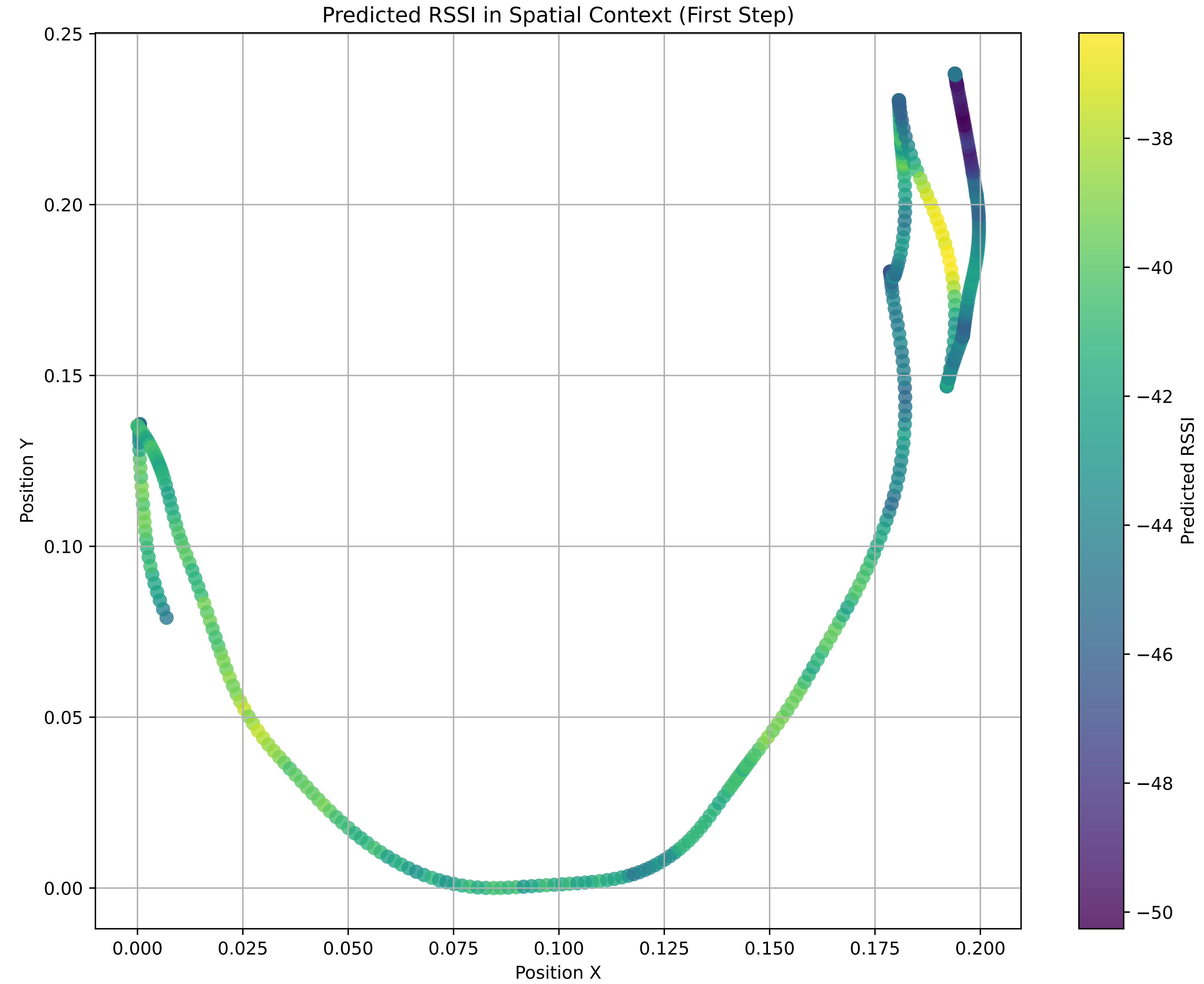


Actual vs Predicted RSSI (First Step)
−37.5
−40.0
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
Predicted RSSI
−54
−52
−50
−48
−46
−44
−42
−40
Actual RSSI

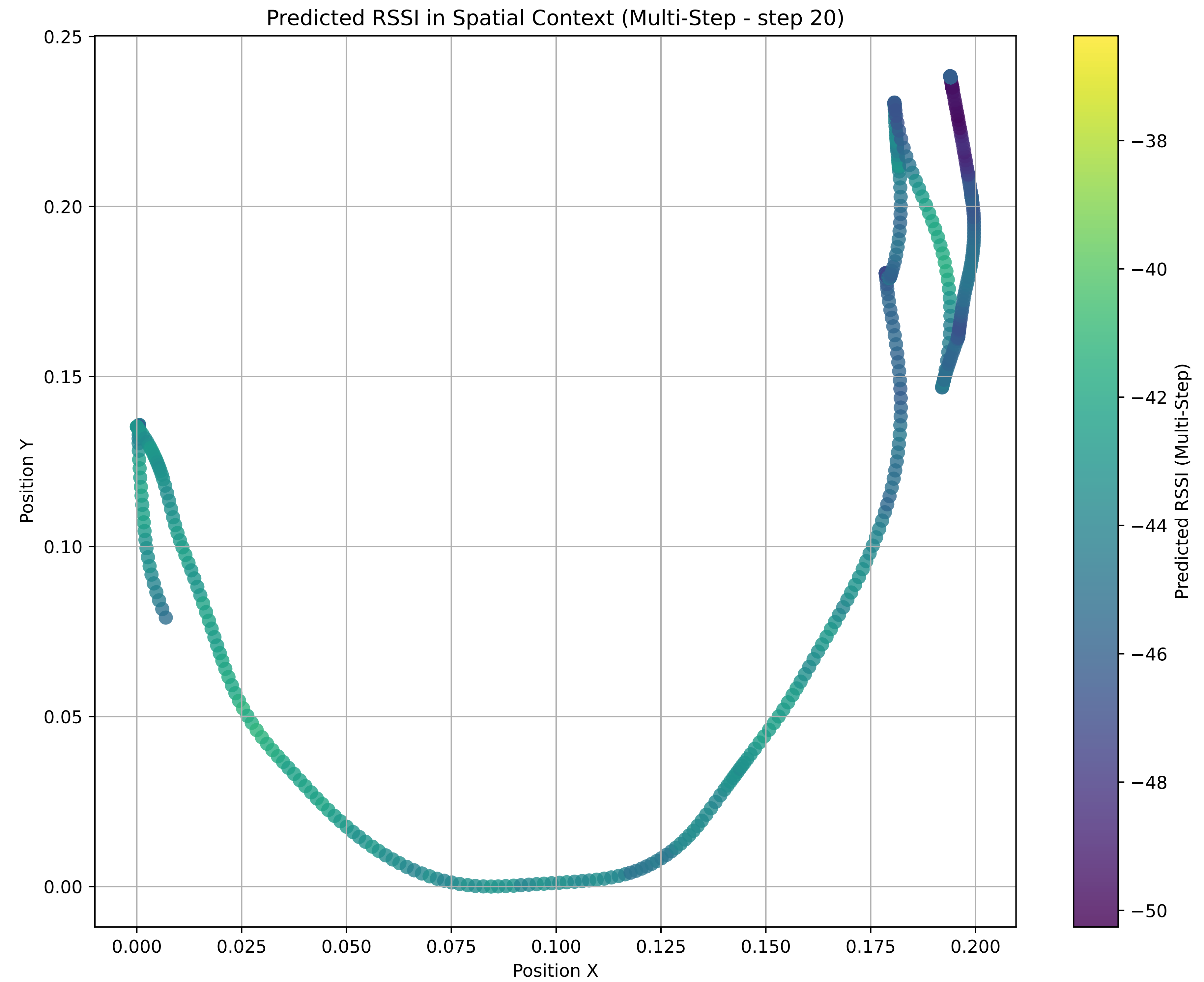
Actual vs Predicted RSSI (Last Step – step 20)
Predicted RSSI
Actual RSSI
−40
−42
−44
−46
−48
−50
−52
−54
−54
−52
−50
−48
−46
−44
−42
−40


Distribution of Prediction Errors (First Step)
Frequency
Error (Actual RSSI - Predicted RSSI)
0
20
40
60
80
100
120
140
160
−8
−6
−4
−2
0

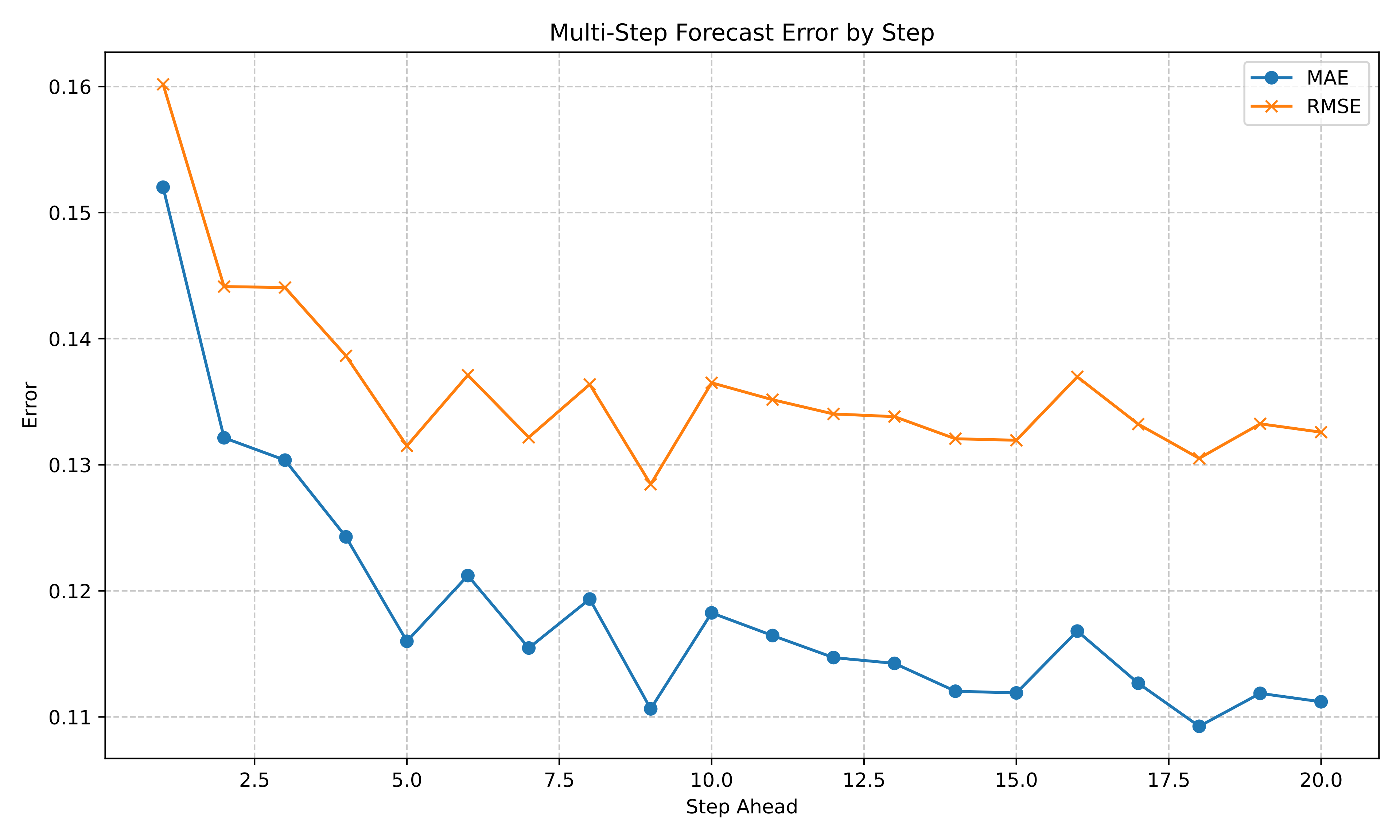

Distribution of Prediction Errors (Last Step – step 20)
Frequency
0
20
40
60
80
100
120
140
160
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Error (Actual RSSI - Predicted RSSI)



Predicted RSSI in Spatial Context (First Step)
Position Y
Position X
Predicted RSSI
0.25
0.20
0.15
0.10
0.05
0.00
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
−38
−40
−42
−44
−46
−48
−50



Predicted RSSI in Spatial Context (Multi-Step - step 20)
0.25
0.20
0.15
0.10
0.05
0.00
Position Y
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Position X
−38
−40
−42
−44
−46
−48
−50
Predicted RSSI (Multi-Step)



Multi-Step Forecast Error by Step
MAE
RMSE
Error
Step Ahead
0.16
0.15
0.14
0.13
0.12
0.11
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

```
Resumo Automático
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Dataset:        rssi_odom_dataset_11apr25.parquet
Modelo:         GRU
Forecast steps: 20

MAE / RMSE (dB)
    • 1º passo : 0.15 / 0.16
    • Mediana  : 0.12 / 0.13
    • 20º passo : 0.11 / 0.13

Crescimento do erro: -26.8%   →   Estável (erro cresce < 20 %)

3 piores passos (MAE): 1 (0.15), 2 (0.13), 3 (0.13)

Recomendação:
O modelo é adequado para decisões de hand-over até o horizonte máximo configurado.
```